

Service Quality and Transit Equity in the NYC Subway System

A line-level analysis of Customer Journey Time Performance, 2015–2026

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May 2026

Executive Summary

Transit equity analysis typically asks whether subway service is worse in neighborhoods with lower-income or more transit-dependent residents. This brief asks the same question of New York's subway over an eleven-year monthly panel (2015–2026), using the MTA's Customer Journey Time Performance metric and an 800-meter walking-buffer demographic match. Line-level service quality varies substantially and persistently — the 1 train remains consistently above the B train across the panel — but this variation does not track service-area median household income (Pearson $r = 0.13$).

Service quality is more clearly organized by capital investment history. The two lines that have received Communications-Based Train Control upgrades run measurably better and more consistently than non-CBTC lines, a descriptively robust association that should not be read as a strict causal estimate. The brief argues that transit equity in New York is better understood through investment and operational decisions, including signaling upgrades and off-peak frequency, than through surrounding demographics alone.

1 Background and Motivation

The New York City subway carries approximately 3.4 million riders a day and remains the principal rail transit system for mobility across the five boroughs. Whether the system delivers comparable service quality to different parts of the city is therefore a question of distributional consequence — not just for transit policy, but for housing access, labor-market participation, and time poverty among workers without alternatives. This is the question that transit equity research has tried to make tractable.

The standard approach pairs service-quality outcomes with neighborhood demographics: are lines that serve lower-income, higher-transit-dependency, or majority-minority communities measurably worse along some dimension of service? Work in environmental justice and transit equity has applied

this framework to bus reliability, subway breakdown rates, station accessibility, and stop spacing, often asking whether burdens fall disproportionately on lower-income communities and communities of color. The framework is intuitive, accessible to non-specialist audiences, and aligned with how city agencies define their own equity obligations.

This brief takes the same starting question — whether subway service quality is distributed equitably across New York — but does so at the line level, over an eleven-year monthly panel, using the MTA's own customer-experience metric (CJTP) rather than train-centric measures like on-time arrival. Two features of this setup change what the analysis can say. First, the long panel exposes structure that single-cross-section studies miss: which lines are persistently good, which are persistently weak, which actually shift over time. Second, the customer-journey framing measures what riders experience — a journey that completes within five minutes of when they were told it would — rather than what trains do at the platform level.

The findings that follow document substantial and persistent line-level differences in service quality, but they also complicate the standard equity reading. Service quality, in this panel, is not organized by the income composition of the surrounding neighborhoods. It is organized by something else, and identifying what that something is — and what it implies for how equity-oriented investment should be allocated — is the central question taken up in the rest of the brief.

2 Data and Method

The analysis combines three data sources. Service-quality data come from the MTA's Customer Journey-Focused Metrics (CJM) table on the New York State Open Data portal, which reports monthly Customer Journey Time Performance (CJTP) and related measures by subway line and time-of-day period from January 2015 through March 2026. CJTP is the share of customer journeys completed within five minutes of scheduled time, estimated by linking fare-payment records with train movement data. The panel covers 21 main lines (excluding the 42nd Street Shuttle), each observed monthly under both peak and off-peak service, yielding 5,624 line-period-month rows.

Demographic data come from the American Community Survey 5-year estimates for 2018–2022, accessed through the Census API. The relevant tract-level variables are median household income (B19013) and means of transportation to work (B08301), which is used to construct transit-commuter weights. Census tracts in the five New York City counties (Bronx, Kings, New York, Queens, Richmond) are included where ACS income and commuting-mode estimates are available; income is reported in 2018–2022 nominal dollars without further inflation adjustment.

To connect line-level service quality to neighborhood demographics, we use the MTA's General Transit Feed Specification (GTFS) static feed to locate each station, consolidate platform-level stops by parent station, and assign each station to its lines. Around each station we draw an 800-meter walking buffer in projected coordinates (EPSG:2263, New York State Plane), dissolve

buffers within a line, and intersect the result with the tract layer. Each tract that overlaps a line's combined service area is weighted by the overlap fraction multiplied by the tract's total transit commuters; the line-level demographic value is the commuter-weighted average across overlapping tracts. This weighting privileges tracts whose residents actually use transit, rather than treating every square meter of buffered area as equivalent.

The regression panel is built from the CJM table by line and month, with B Division lines (B, D, F, M, and the lettered Brooklyn–Manhattan lines) appearing from 2017 onward and IRT lines (1 through 7) present for the full panel. Two-way fixed-effects estimation is implemented in `linearmodels.PanelOLS` with entity (line) and time (month) effects and standard errors clustered at the entity level. The CBTC event-study specification replaces the `cbtc_active` dummy with a set of binned event-time indicators (pre-25-plus, pre-24-19, pre-18-13, pre-12-7, pre-6-1 as reference, post-0-5, post-6-11, post-12-23, post-24-47, post-48-plus); coefficients are interpreted relative to the omitted pre-6-1 window. The two CBTC activation dates used are February 2012 for the L and November 2018 for the 7. All analysis is implemented in Python (pandas, geopandas, linearmodels, matplotlib); replication code is available at the repository linked at the end of this brief.

3 Findings

3.1 Cross-line differences are systematic, not noise

Variation in subway service quality might be noise — month-to-month fluctuation averaging out across a complex system. Eleven years of monthly Customer Journey Time Performance (CJTP) data argue otherwise. Across 21 lines over eleven years, line identity dominates: the 1 train's CJTP never falls below 0.87 in any month of the panel; the B train's never rises above 0.80. That gap is wider than any seasonal swing, any winter storm, any service-change event in the window.

The pattern is not where casual riders might place it. The 4/5 Lex express, often imagined as the system's flagship, sits in the worst tier through 2015–2017 (CJTP $\approx$ 0.73–0.77), recovering only modestly after. Outer-borough lines like the 7 and the L — frequently characterized as overcrowded and unreliable — are among the steadiest performers. A formal panel test confirms the visual reading. An F-test for poolability rejects the idea that the 21 lines can be treated as sharing a common service-quality level once month effects are accounted for ($F = 55.5$, $p < 0.001$).

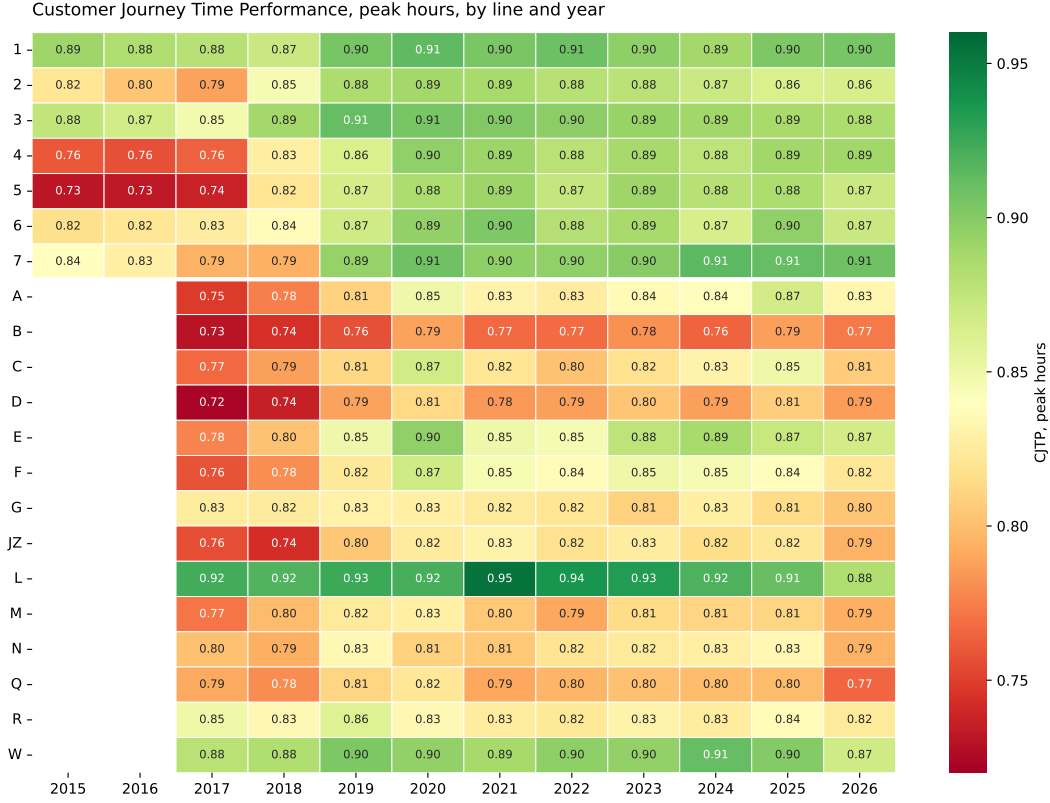


Figure 1: Customer Journey Time Performance by line and year, peak hours, 2015–2026. Each row is a subway line, each column a year; color encodes the annual mean CJTP. Off-peak service is analyzed separately in §3.3.

The horizontal banding — solid colors holding across columns — shows that service quality is organized by line, not just by time.

3.2 The pandemic shock moved the metric, not just the service

The spring 2020 lockdown did not produce a simple deterioration in measured subway service quality. Systemwide CJTP dipped in April 2020, then rebounded in May and June to levels above the immediate pre-lockdown months. This is not a recovery story. It is a denominator story.

The explanation lies in how the metric is built, not in a sudden improvement in operations. CJTP is a ratio: customer journeys completed within five minutes of schedule, divided by total customer journeys measured. When ridership collapsed and the MTA thinned the schedule in parallel, both the denominator and the operating environment changed. Trains that did run were less crowded, dwelled less at stations, and met a sparser schedule that was easier to keep. Apparent service improvement can therefore reflect the metric responding to the shock, not the system performing better for everyday riders.

The pattern was uneven across lines. The 7 train shows it most starkly: peak-period CJTP reached 0.96 in May 2020, well above its 2019 baseline near 0.89. Other lines moved less, and the L train, often discussed alongside the 7 because both run on Communications-Based Train Control, did not show the same lockdown spike. Whatever happened in April–May 2020 was not a uniform systemwide improvement; it varied sharply by line and period.

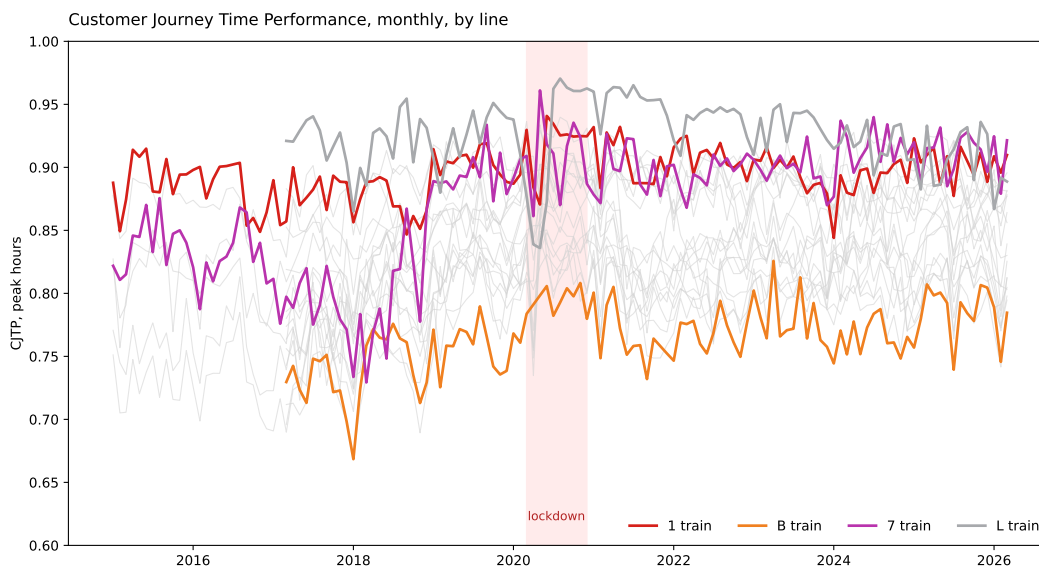


Figure 2: Customer Journey Time Performance by line, January 2015 – March 2026. The spring 2020 lockdown is shaded. Selected lines are highlighted: the 1 (red), B (orange), 7 (purple), and L (gray). The aggregate pattern shows an April 2020 dip followed by a May–June rebound; among the highlighted lines, the 7 shows the clearest May 2020 spike.

The 2020 episode is a useful caution: ratio-based service metrics can move in counterintuitive directions when the underlying flows are disrupted. For that reason, the analysis that follows emphasizes line-level persistence and within-line changes rather than aggregate movement alone.

3.3 Off-peak service quality is a frequency problem

Off-peak hours are often imagined as the subway’s calmer state: fewer riders, less crowding, fewer dwell-time delays. By the CJTP measure, they are not. Across the 21 lines studied, peak-period CJTP is systematically higher than off-peak CJTP, by an average of 2 to 5 percentage points. The penalty is largest on long outer-borough lines: the F, A, and E each lose roughly 5 percentage points moving from peak to off-peak service.

The mechanism is not crowding. Off-peak trains are emptier and faster between stops. The mechanism is headway. Off-peak schedules are thinner — trains run every 12 to 20 minutes rather than every 4 to 6 — so a single missed train, a single signal hold, or a single dispatching delay

translates into a wait long enough to push the journey past the five-minute on-time threshold. CJTP is, in this sense, sensitive to schedule density: when service is sparse, small operational frictions become large rider-experienced delays.

Two lines make the headway interpretation visible from the other direction. The G train is the clearest reversal, with off-peak CJTP about 1 percentage point higher than peak. The 6 is essentially flat, with almost no off-peak penalty. Both are short, self-contained lines that do not depend on the Manhattan trunk for connections, and both maintain relatively dense off-peak headways. When off-peak service is not thinned, the off-peak penalty disappears.

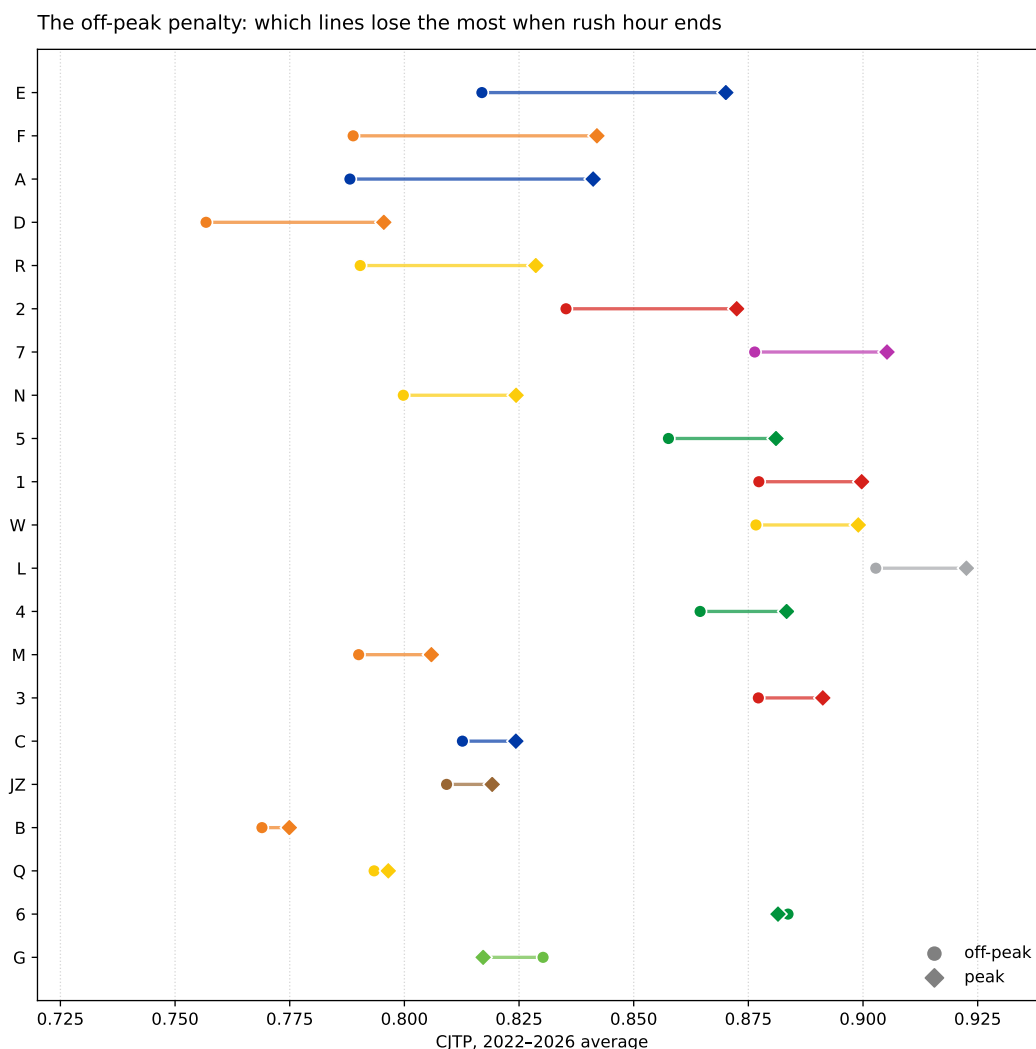


Figure 3: Peak versus off-peak Customer Journey Time Performance by line, panel-period mean. Each row shows one line; the gap between the two markers is the off-peak penalty. Long outer-borough lines (F, A, E) show the largest penalties; the G train reverses the pattern and the 6 train shows essentially no penalty.

The implication is that off-peak service quality is set by scheduling decisions, not by ridership demand. Lines that lose the most quality off-peak are the lines whose off-peak schedules have been thinned most aggressively. This raises a distributional question taken up in §4: which riders rely on off-peak service, and which lines make off-peak travel costly?

3.4 Service quality does not track neighborhood income

A standard transit-equity question asks whether subway lines that serve lower-income neighborhoods provide worse service. To test this directly, we paired each line with its transit-commuter-weighted service-area median household income. The measured association is weak and not statistically distinguishable from zero. The Pearson correlation between line-level mean peak CJTP and service-area median income is 0.13; with only 21 lines, its 95% confidence interval runs from about -0.32 to $+0.53$ — wide enough to contain moderate associations in either direction. A line-month OLS regression is consistent with this: the coefficient is 0.010 per \$10,000 of service-area income ($p = 0.18$). The disciplined reading is not that service quality is unrelated to neighborhood income, but that this line-level test cannot detect an income gradient at the sample size available.

The geography behind the null is informative. The lowest-income service areas in the system belong to outer-borough working-class corridors: the D (\$87k), JZ (\$92k), 7 (\$93k), and B (\$95k). The highest-income areas belong to Manhattan-anchored or Brooklyn-trunk lines: the N (\$129k), W (\$126k), Q (\$125k), and R (\$123k). These two groups do not differ systematically in CJTP. Some low-income lines (the 7 in particular) outperform high-income ones; some high-income lines (the Q, R, and N during much of the panel) underperform low-income ones. The line-by-line scatter shows no organizing slope.

The most informative cases are the lines that income fails to explain. The 7 train serves a service area near \$93k yet sits in the top quartile of CJTP, with peak values around 0.91 in recent years. The L train serves a \$111k area and posts peak CJTP near 0.92. Both run on Communications-Based Train Control, a major signaling upgrade examined in §3.5. The B and the D, by contrast, sit at the low end of CJTP regardless of how their service-area income compares with the rest of the system. Whatever explains line-level service quality, service-area income does not capture it in this comparison.

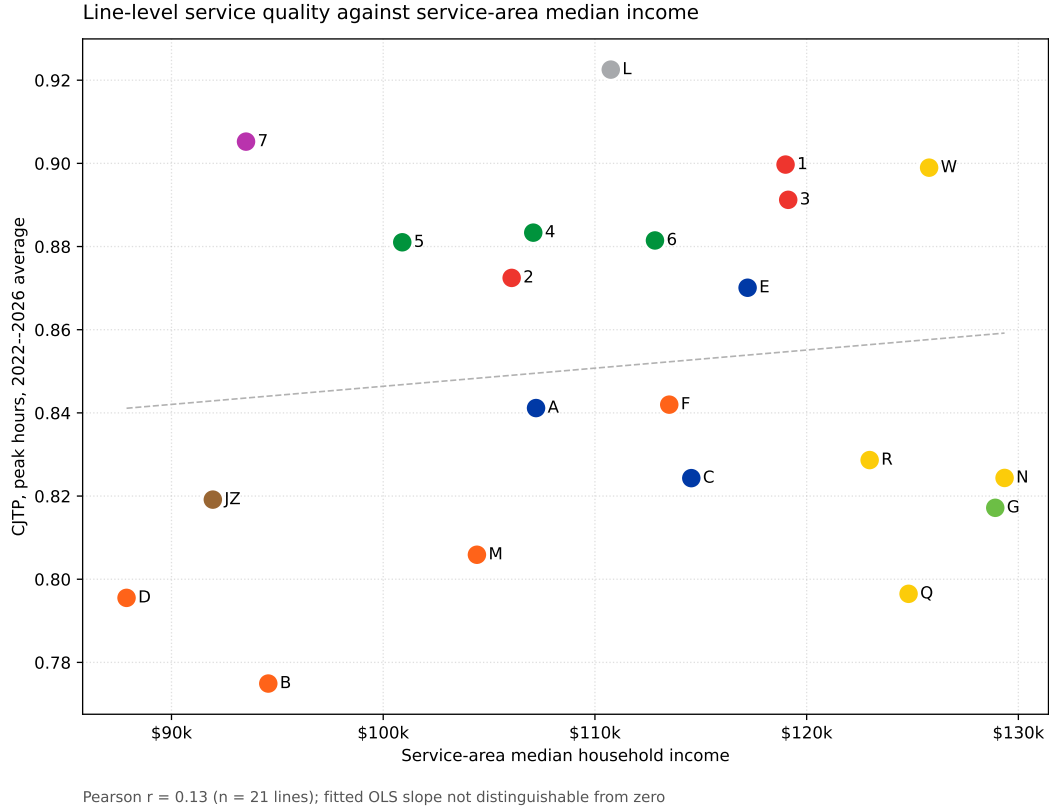


Figure 4: Line-level mean peak Customer Journey Time Performance, 2022–2026, plotted against service-area median household income (ACS 2018–2022, weighted by commuter volume within an 800-meter walking buffer of each line’s stations). The fitted line is OLS; its slope is not statistically distinguishable from zero. The two CBTC lines (the 7 and L) sit well above the fit.

The inability to detect an income gradient does not weaken the equity question; it redirects it. The failure to find a relationship at this sample size does not establish that none exists — the confidence interval above is wide, and §5 notes that line-level aggregation may itself mask a within-line gradient — but it does mean income cannot be shown to be the axis along which service quality is organized here. The line-level variation documented in §3.1 is real, persistent, and consequential for riders. §3.5 takes up a candidate the data can speak to more directly: capital investment history.

3.5 CBTC investment is associated with persistent service gains

Communications-Based Train Control (CBTC) replaces the subway's century-old fixed-block signaling with a system that tracks each train's position continuously and allows trains to operate closer together at higher reliability. The MTA has completed CBTC on two lines in the panel window: the L (operational by February 2012, before the panel begins) and the 7 (completed in November 2018). The lines that have it run differently from the lines that don't.

A two-way fixed-effects panel regression, with line and month effects and standard errors clustered at the line level, estimates that CJTP on a CBTC-active line is 3.0 percentage points higher than on a non-CBTC line in the same month ($p = 0.017$; 95% CI: 0.6 to 5.5 percentage points). The estimate is sizable in the context of a metric whose line-level standard deviation is roughly 4 percentage points.

The event-study version of the same regression complicates the causal reading without overturning the descriptive one. Pre-period coefficients (24 to 1 month before activation) fluctuate between +6.9 and -2.4 percentage points, drifting rather than holding flat. With only the 7 train providing within-panel CBTC variation — the L was already CBTC-active at the start of the data — pre-trends are identified off a single line, and the parallel-trends assumption that licenses a strict causal interpretation cannot be defended here. Post-activation coefficients remain positive, generally in the +5 to +7 percentage-point range, and the treated line appears less volatile month to month.

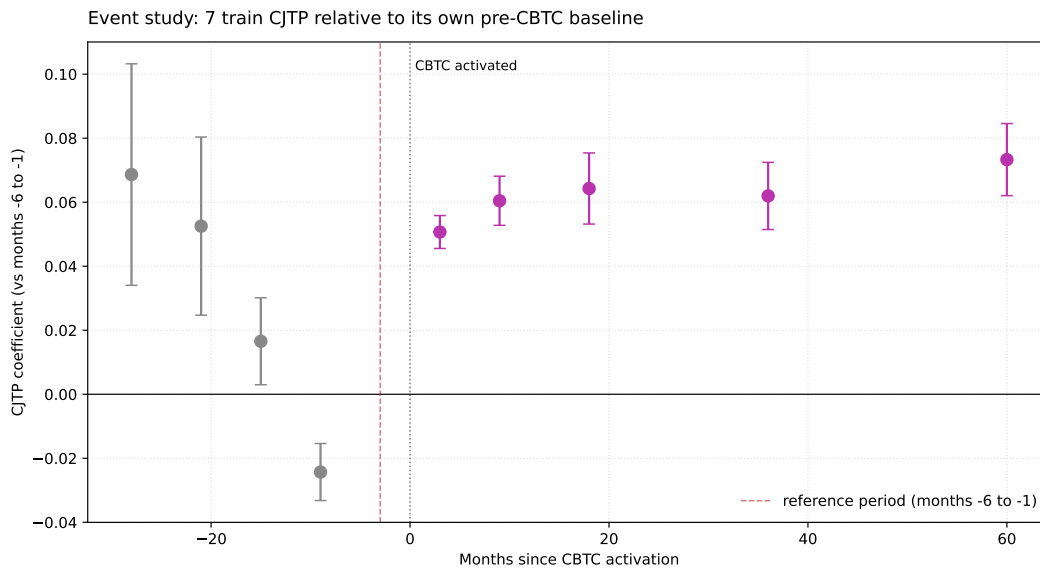


Figure 5: Event study of Customer Journey Time Performance around CBTC activation. Each point is a coefficient from a two-way fixed-effects regression with line and month effects; vertical bars show 95% confidence intervals from line-clustered standard errors. The L train (activated February 2012) and the 7 train (activated November 2018) are the treated lines; only the 7 provides within-panel identification.

Two things follow. First, CBTC lines do not just sit at a higher CJTP level — they sit there with an apparent reduction in month-to-month variance, which is the rider-relevant attribute that capacity-only metrics miss. Second, this analysis cannot rule out that some other line-specific change coincided with CBTC activation, particularly on the 7. With only two treated lines and one providing identification, single-line confounders (a supervisor, a fleet upgrade, a track program) are observationally indistinguishable from the signaling system itself. The descriptive claim — CBTC

lines run measurably better and more consistently than non-CBTC lines — is well supported. The mechanistic claim — that signaling specifically caused the gain — requires more lines, more variation, or both. The MTA's planned CBTC expansion on additional corridors will provide more variation for future evaluation. Until then, the existing two-line evidence points to investment history as a more plausible equity mechanism than neighborhood income alone.

4 Implications for Transit Equity Policy

The findings carry three implications for how transit equity is operationalized in New York.

First, line-level service quality varies systematically across the system, and the variation is not noise. The cross-line gap documented in §3.1 — eleven years in which the 1 train and the B train never overlap on CJTP — is large enough to materially affect a rider's commute and persistent enough to be a feature of the system rather than a transient condition. Any equity diagnostic that aggregates service across lines, or that treats month-to-month variation as the relevant unit of analysis, will miss the structure that riders actually experience.

Second, the income-based equity diagnostic does not turn up a detectable gradient in this panel. Line-level service quality shows no measurable association with service-area median household income (§3.4), and the lines that most clearly outperform what their income would predict are not random: they are the lines that have received signaling investment (§3.5). The takeaway is not that income is irrelevant to transit equity. It is that line-level service quality in New York is more clearly organized by capital investment history than by the demographic composition of surrounding neighborhoods. An equity audit that examines only demographic correlates risks missing the mechanism through which inequality is reproduced — or ameliorated — in actual service delivery.

Third, off-peak service quality (§3.3) is a distributional question that the standard peak-focused service framework does not surface. Off-peak headways are scheduling decisions, and the lines that suffer most when off-peak service is thinned are not the lines whose riders have the most alternatives. Asking who relies on off-peak service, and on which lines, is a tractable next step that the data in this brief do not directly answer but plainly invite.

Taken together, these implications reframe rather than abandon the transit equity question. The distributional concern is real; the appropriate axis along which to evaluate it is investment history and operational decisions, not surrounding demographics alone. For an agency setting capital priorities or evaluating service plans, this means treating the CBTC rollout schedule, the off-peak frequency table, and the next-line investment decision as equity-relevant choices — not because they target specific communities, but because they shape which lines deliver consistent service to the riders who depend on them.

5 Limitations

Several limitations qualify the findings in this brief.

The CBTC analysis identifies its effect off a single line. Only the 7 train transitions into CBTC during the panel window; the L is already treated at the start of the data. As discussed in §3.5, this means single-line confounders — a maintenance program, a fleet replacement, a supervisory change — cannot be separated from the signaling system itself. The estimated post-activation lift is descriptively well supported but is not a credible point estimate of the causal effect of CBTC specifically. The MTA's planned CBTC expansion will eventually permit stronger identification.

Service-area demographics are constructed from ACS 2018–2022 5-year estimates and treated as static throughout the eleven-year panel. Neighborhood composition along several subway corridors has shifted meaningfully over the panel window, particularly in parts of Brooklyn and Queens. The income figures used in §3.4 are therefore best understood as a recent snapshot rather than as the demographic conditions under which the full 2015–2026 service record was produced.

The income null in §3.4 is also sensitive to the scale at which it is measured. Assigning a single service-area income to each line averages over every neighborhood the line passes through, and several lines span an enormous internal range — the 2 train, for example, serves both the Upper West Side and Brownsville. The census tracts underlying this analysis run from roughly \$11{,}000 to the ACS \$250{,}000 top-code in median household income, yet the commuter-weighted line-level figures compress into a narrow \$88k–\$129k band. A relationship between income and service quality that operates within lines, or across stations, could be averaged out entirely once the unit of analysis is the line. The §3.4 result is therefore best read as the absence of a *line-level* income gradient, not as evidence that income and service quality are unrelated at finer geographies; a station- or segment-level analysis could reach a different conclusion.

The 800-meter walking buffer used to assign tracts to lines is a reasonable but imperfect proxy for who a line serves. Riders do not in fact arrive only on foot; many transfer from buses, drive to park-and-rides, or walk longer distances. A more refined service-area definition would incorporate access mode and travel-time isochrones rather than a single distance threshold.

CJTP is rider-centric but not all-encompassing. It captures schedule adherence but does not directly measure crowding, comfort, safety, accessibility for riders with disabilities, or the availability of service late at night. A line that performs well on CJTP can still be deficient on dimensions that matter for equity. The findings here therefore characterize one component of service quality, not the totality of it.

Finally, the brief examines service quality but not its upstream determinants. Why some lines have received signaling investment while others have not, why off-peak schedules are thinned on particular corridors, and how capital decisions are made at MTA are political-economic questions that the data here can describe the consequences of but not explain. A complete equity analysis

would integrate this descriptive evidence with the institutional history of capital prioritization, which is beyond the scope of a brief of this length.

Replication materials, including data, code, and figures, are available at <https://github.com/yidan114/please-stand-clear-of-the-closing-data>.